

FA-KGD: Frequency-Aware Kalman-Guided Diffusion for Accelerated MRI Reconstruction

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Abstract

*Diffusion priors for accelerated MRI reuse one pre-trained model across masks and inverse problems, but existing samplers treat k-space measurement noise as isotropic and enforce data consistency uniformly at every denoising step. We propose **Frequency-Aware Kalman-Guided Diffusion (FA-KGD)**, a training-free sampler that adds no learned weights to a frozen EDM prior pretrained on fastMRI brain: it replaces the scalar measurement noise with a frequency-dependent Kalman gain and admits high-frequency measurements through a progressive k-space radius schedule, only after the iterate has sharpened enough to support them. Our central finding is a scaling law. On a 192-slice, 12-volume fastMRI brain sweep at $R=4$, the Π GDM baseline plateaus across $T \in \{20, 50\}$ then regresses at $T=100$, while FA-KGD's PSNR gain grows monotonically with T (+0.11/ +0.23/ +0.39/+**0.51** dB at $T=10/20/50/100$), positive on every one of the 12 volumes. The pattern reproduces cross-domain on 192 single-coil knee slices (+0.16 $\rightarrow$ +0.40 dB across the same T sweep at $R=4$), confirming the effect is not specific to the brain data the prior was trained on. We frame these gains as sampler improvements, which could translate into clinical benefits if thoroughly validated by radiologists.*

1. Introduction

Magnetic Resonance Imaging (MRI) offers exceptional soft-tissue contrast but long acquisition times bottleneck its clinical use. *Accelerated MRI* recovers an image from a fraction of k-space samples: at $4\times$ acceleration (equivalently $R=4$, the notation we use throughout) a scan is 75% shorter, which can reduce motion artifacts, enable pediatric and cardiac protocols, and raise scanner throughput. Mathematically, undersampling removes Fourier measurements and leaves a severely underdetermined reconstruction problem; we introduce the precise linear model in Section 3.

Diffusion models [1–3] have emerged as compelling priors for this task, combining expressive learned image statistics with principled posterior sampling [4, 5]. Super-

vised end-to-end methods such as the End-to-End Variational Network (E2E-VarNet) [6] still hold the fastMRI leaderboard, but require retraining for every anatomy, coil array, and acceleration factor. Sampler-training-free diffusion methods (Denoising Diffusion Restoration Models (DDRM) [7], Diffusion Posterior Sampling (DPS) [8], Π GDM [9]) trade some absolute quality for generality: one unconditional prior handles arbitrary linear inverse problems across anatomies and acceleration factors without fine-tuning. At the k-space noise regime of real MRI, however, they rely on assumptions that do not always hold in practice.

MRI-specific failure modes. (1) *Isotropic noise:* all assume $\mathbf{R} = \sigma_y^2 \mathbf{I}$, yet k-space noise power varies by 1–2 orders of magnitude between the fully-sampled centre (the auto-calibration signal (ACS) region we reuse for calibration) and the periphery, and again between coils. (2) *DC shock:* uniform data consistency (DC) at every step forces high-frequency residuals to zero while their noise still dominates, creating a distributional mismatch that can inject Gibbs-like ringing. (3) *Open-loop gain:* gains are calibrated once before inference, so any miscalibration propagates through all 100+ steps with no feedback.

Contributions. We introduce FA-KGD, a sampler-training-free method that attacks all three errors with a single, minimal-overhead update:

1. A *self-calibrated frequency-adaptive Kalman gain* derived in closed form, $K_i(t) = \sigma_t^2 / (\sigma_t^2 + \hat{\sigma}_i^2)$, with $\hat{\sigma}_i^2$ estimated from the ACS lines already in the acquisition, with no side-channel calibration scan.
2. *Frequency-Progressive Data Consistency (FPDC):* a k-space radius schedule $r(t)$ that enforces DC only on frequencies compatible with the current iterate's noise level, eliminating DC shock.
3. A γ -family analysis of online noise estimation showing why the naive bias correction ($\gamma=1$) can severely overcorrect when denoiser error is small. We use $\gamma=0$ as a conservative, stable default, without claiming it is a universal optimal value.

We show preliminary evidence of *superadditive synergy*:

FPDC and the adaptive gain are mutually enabling rather than independently additive (+1.9 dB jointly vs. 0 dB + 0.7 dB separately on an oracle denoiser).

Input/output. Given undersampled k-space $\mathbf{y} \in \mathbb{C}^m$ (variable-density Cartesian mask Ω with a fully-sampled 24×24 ACS region) and a frozen pretrained EDM score network μ_θ [10], FA-KGD runs a T -step DDIM reverse process applying its per-frequency Kalman gain and frequency-progressive data consistency at every step (Figure 1 and Algorithm 1) to output a complex image x_0 , evaluated as $|x_0|$ against the fully-sampled ground truth via PSNR and SSIM (defined in Section 6).

2. Background and Related Work

2.1. Score-Based Diffusion Models

The Denoising Diffusion Probabilistic Model (DDPM) [1] defines a forward noising process $q(x_t | x_0) = \mathcal{N}(\sqrt{\alpha_t}x_0, \sigma_t^2\mathbf{I})$ and trains a score network $s_\theta(x_t, t) \approx \nabla_{x_t} \log p(x_t)$ to reverse it, generalised to continuous-time stochastic differential equations (SDEs) by Song et al. [2] and to the practically dominant Elucidating Diffusion Model (EDM) parameterisation by Karras et al. [3]. At inference, each step produces a Tweedie-denoised estimate $\mu_\theta(x_t) \triangleq \mathbb{E}[x_0 | x_t]$ that we treat as the “prior” side of a measurement-prior fusion.

2.2. Diffusion Samplers for Inverse Problems

DDRM [7] uses the singular value decomposition (SVD) of the forward operator and solves noisy linear inverse problems in ~ 20 steps. DPS [8] back-propagates a Laplace-approximated likelihood gradient through the Tweedie estimate. IIGDM [9], our primary baseline, derives the exact posterior update for the linear-Gaussian model:

$$\hat{x}_{t-1} = \mu_\theta(x_t) + r_t \mathbf{F}_\Omega^H \left(\mathbf{F}_\Omega \mathbf{F}_\Omega^H + \frac{\sigma_y^2}{\sigma_t^2} \mathbf{I} \right)^{-1} (\mathbf{y} - \mathbf{F}_\Omega \mu_\theta(x_t)), \quad (1)$$

where $\mathbf{F}_\Omega \in \mathbb{C}^{m \times n}$ is the subsampled Fourier operator (the Fourier transform restricted to the sampled set Ω), $\mathbf{y}$ the undersampled k-space measurements, σ_y^2 the scalar measurement-noise variance, and r_t a step-size weight (σ_t and μ_θ as above). This is a Kalman step with *isotropic* covariances applied uniformly to every measured k-space location. Ambient Diffusion Posterior Sampling (ADPS) [10] extends the family to ambient-trained models via annealed Langevin likelihood guidance; its default schedule targets $T \geq 300$ and underperforms at the low step budgets we evaluate.

2.3. Diffusion Priors for MRI and Covariance Estimation

Jalal et al. [4] first demonstrated score-based compressed-sensing MRI with a posterior-sampling

formulation, and Chung & Ye [5] extended it to the multi-coil fastMRI setting; both still assume isotropic measurement noise and enforce DC at every step across the full sampled mask. Peng et al. [11] improve *image*-domain posterior covariance using a calibration dataset; FA-KGD instead estimates *measurement*-domain (k-space) covariance from the scan’s own ACS lines, with no additional data. Fast Diffusion EM [12] jointly estimates image and degradation, but its M-step is a per-step maximum a posteriori (MAP) optimisation; ours is a single element-wise update. Neither work addresses DC shock or characterises how denoiser quality governs bias correction in online noise estimation.

2.4. Parallel MRI, Datasets, and the Supervised Gap

ESPIRiT [13] is the canonical compressed-sensing baseline for parallel MRI, estimating coil sensitivities from ACS data. fastMRI [14] is the standard benchmark; we use the EDM checkpoints from Ambient Diffusion [10] pre-trained on fully-sampled fastMRI brain without modification. Supervised models (U-Net [14], E2E-VarNet [6]) still define the fastMRI state of the art by jointly learning reconstruction and coil sensitivity maps over all $N_c \approx 16$ coils. The ~ 10 dB gap in Table 2 reflects this information-access difference (multicoil raw data vs. a single root-sum-of-squares (RSS) combined channel) as much as model capacity; it is orthogonal to the sampler improvements studied here.

3. Problem Formulation

3.1. MRI as a Linear Inverse Problem

An MRI scanner measures complex-valued Fourier coefficients of an image $x \in \mathbb{C}^n$:

$$\mathbf{y} = \mathbf{F}_\Omega x + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \mathbf{R}), \quad (2)$$

where $\mathbf{F}_\Omega \in \mathbb{C}^{m \times n}$ is the subsampled Fourier operator ($m \ll n$) and $\mathbf{R}$ is the measurement noise covariance. We use $\mathbf{F}_\Omega$ throughout; the FPDC variant in Section 4 generalises it to the time-varying operator $\mathbf{F}_{\Omega(t)}$. Accelerated MRI sets $m/n \in \{1/4, 1/8\}$ (equivalently $R=4$ and $R=8$), reducing scan time by the same factor at the cost of an underdetermined reconstruction.

Experimental object. The public EDM checkpoint accepts a two-channel complex image, not raw multi-coil k-space, so we use a single coil-combined complex image with the single-coil operator $\mathbf{F}_\Omega = \mathbf{M}\mathbf{F}$ and compute PSNR/SSIM on magnitude/RSS-style targets. The symbols $x, \mathbf{y}, \mathbf{F}_\Omega, \mathbf{R}$ in Equation (2) thus denote the effective coil-combined inverse problem, not a full SENSE forward model; extending FA-KGD to MFS with explicit sensitivity maps $\mathbf{S}$ is future work.

Table 1: How FA-KGD modifies the posterior-sampling update used by prior training-free samplers.

Choice	Prior samplers	FA-KGD
Measurement noise	scalar σ_y^2	frequency map $\hat{\sigma}_i^2$
DC mask	fixed Ω	progressive $\Omega(t)$
Sampling gain	fixed/open loop	residual-refined, clamped
Learned parameters	none added	none added

4. Proposed Method: FA-KGD

FA-KGD combines three components at each denoising step t : (1) ACS-calibrated frequency-adaptive Kalman gain, (2) Frequency-Progressive Data Consistency (FPDC), and (3) online noise refinement with the γ -corrected maximization step (M-step) of an expectation-maximization (EM) update. Figure 1 sketches the per-step flow; Table 1 summarises how these components change the posterior-sampling update relative to prior training-free samplers, and Algorithm 1 presents the full method.

4.1. Component 1 | Self-Calibrated Frequency-Adaptive Gain

Variable-density undersampling masks fully sample a central ACS region (typically 24×24 lines). In the idealized multi-coil setting, one can write a per-frequency replicate variance as

$$\tilde{\sigma}_i^2 = \frac{1}{N_c - 1} \sum_{c=1}^{N_c} |y_{i,c} - \bar{y}_i|^2, \quad (3)$$

where $\bar{y}_i$ is the coil-averaged measurement, N_c is the number of receive coils, and $y_{i,c}$ is the measurement at frequency i in coil c . Equation (3) is a heuristic calibration family rather than a rigorous multi-coil noise estimator (coil differences also contain sensitivity- and object-dependent signal); for our multi-coil experiments we estimate a white noise level from signal-poor image-domain background patches, and for the frequency-dependent experiments we use a controlled synthetic radial noise profile so the ground-truth covariance is known. The estimator family itself, not the sample size, is the dominant source of uncertainty here; we list this as a limitation in Section 7.

Analytic radial noise model. To study the effect of frequency-varying noise in a controlled way, we parameterise the covariance by radial frequency $r = \|\omega\|$, loosely motivated by g-factor profiles in parallel MRI [13], as a low-order radial polynomial,

$$\hat{\sigma}^2(r; \theta) = \theta_0 + \sum_{k=1}^K \theta_k (r/r_{\max})^{2k}, \quad (4)$$

and fit $\theta \in \mathbb{R}^{K+1}$ ($K=2$ throughout) by weighted least squares against the ACS samples Equation (3), with weights

$\propto N_c - 1$. Frequencies outside the ACS are then evaluated by plugging r into Equation (4). The $\hat{\sigma}_i^2$ map is a closed-form function of three numbers per volume; no separate calibration scan is required and the procedure runs in milliseconds. Fitting $O(1)$ parameters against $O(N_z |\text{ACS}|)$ replicates makes the estimator statistically over-determined; the practical engineering trade-off is choosing the parametric family, not collecting more data.

Substituting $\hat{\mathbf{R}} = \text{diag}(\hat{\sigma}_1^2, \dots, \hat{\sigma}_m^2)$ for $\sigma_y^2 \mathbf{I}$ in Equation (1), the Kalman gain reduces to a per-frequency scalar:

$$K_i(t) = \frac{\sigma_t^2}{\sigma_t^2 + \hat{\sigma}_i^2}. \quad (5)$$

The corrected update is:

$$\hat{x}_{t-1}^{\mathbf{K}} = \mu_\theta(x_t) + \mathbf{F}_\Omega^{\mathbf{H}} \mathbf{K}(t) (\mathbf{y}_\Omega - \mathbf{F}_\Omega \mu_\theta(x_t)). \quad (6)$$

When $\sigma_t \gg \hat{\sigma}_i$ (early steps, low-noise frequencies): $K_i \rightarrow 1$. When $\sigma_t \ll \hat{\sigma}_i$ (late steps, noisy frequencies): $K_i \rightarrow 0$. Equation (3) treats every slice independently; Section 4.4 pools across the slice dimension of the same volume to multiply the effective degrees of freedom by N_z at zero additional cost.

4.2. Component 2 | Frequency-Progressive Data Consistency (FPDC)

DC shock. Standard DC enforces hard constraints on all measured k-space frequencies at every step. At step t , the denoiser expects x_t to carry noise level σ_t ; hard DC forces the residual to zero locally, producing a distributional mismatch that can manifest as Gibbs-like ringing in the subsequent step. We do not isolate the ringing artifact visually; the per-band k-space analysis in Table 5 and the edge-localised gains in Figure 2 are the evidence we report for this mechanism.

Fix. FPDC enforces DC only within a frequency radius $r(t)$ that expands as denoising progresses:

$$r(t) = r_{\text{ACS}} + (r_{\max} - r_{\text{ACS}}) \cdot \left(1 - \frac{t}{T}\right)^\beta, \quad (7)$$

where r_{ACS} is the ACS radius, $r_{\max}$ is the full k-space radius, and $\beta > 0$ controls the expansion rate. The joint update is:

$$\hat{x}_{t-1} = \mu_\theta(x_t) + \mathbf{F}_{\Omega(t)}^{\mathbf{H}} \mathbf{K}(t) (\mathbf{y}_{\Omega(t)} - \mathbf{F}_{\Omega(t)} \mu_\theta(x_t)), \quad (8)$$

where $\Omega(t) = \{i \in \Omega : \|\omega_i\| \leq r(t)\}$. This low-to-high schedule is an inductive bias rather than an optimality claim. It matches the empirical tendency of diffusion reconstructions to recover coarse structure before fine detail, but future schedules could tie $r(t)$ directly to σ_t , an estimated image power spectrum, or a frequency-wise signal-to-noise ratio.

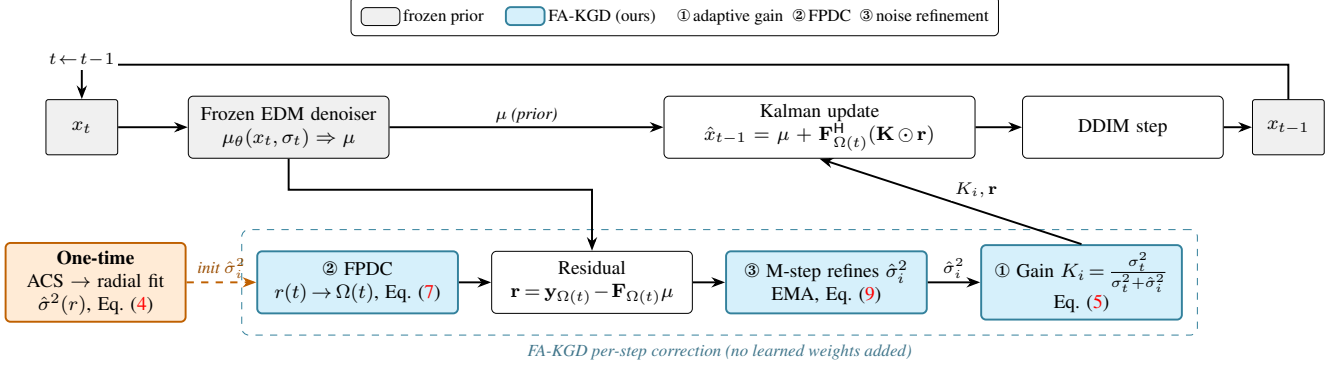


Figure 1: FA-KGD per-step pipeline. A frozen EDM denoiser (grey) produces the prior estimate μ ; FA-KGD’s three training-free components (blue) then form a frequency-adaptive, progressively-gated Kalman correction before the DDIM step. The per-frequency noise map $\hat{\sigma}_i^2$ is initialised once from an ACS radial fit (Equation (4)) and refined online by the M-step (Equation (9)). No weights are added to the frozen prior.

4.3. Component 3 | Online Noise Refinement and the γ -Family

After computing $\mu_\theta(x_t)$, we update $\hat{\sigma}_i^2$ via:

$$\hat{\sigma}_i^2 \leftarrow \alpha(t) \hat{\sigma}_i^2 + (1 - \alpha(t)) \max(|r_i^{(t)}|^2 - \gamma \sigma_t^2, \epsilon), \quad (9)$$

where $r_i^{(t)} = y_i - [\mathbf{F}_{\Omega(t)} \mu_\theta(x_t)]_i$ and $\alpha(t) = 1 - (1 - \alpha_0)(\sigma_t / \sigma_T)$ is a time-decaying exponential moving average (EMA) momentum.

The γ -family. As a heuristic model, suppose a well-trained score network has denoiser error fraction $\eta^2 \ll 1$ and that denoiser error is additive, isotropic, and uncorrelated with measurement noise. Then the expected squared residual decomposes as

$$\mathbb{E}[|r_i^{(t)}|^2] = \hat{\sigma}_i^2 + \eta^2 \sigma_t^2, \quad (10)$$

i.e. true measurement noise *plus* a small, diffusion-driven residual that vanishes with t . Under these assumptions, the naive bias correction $\gamma=1$ subtracts the full σ_t^2 , over-correcting by $1/\eta^2$ and driving $\hat{\sigma}_i^2 \rightarrow 0$; the gain then collapses to $K_i \rightarrow 1$ everywhere and DC is enforced without regularisation. In simulation $\gamma=1$ degrades PSNR by 12.8 dB relative to IIGDM, while $\gamma=0$ yields +0.7 dB. We therefore use $\gamma=0$ as a conservative default for strong denoisers (Section 7).

4.4. FA-KGD-3D: Volumetric ACS Pooling

MRI volumes are 3D, yet FA-KGD as written treats each slice independently. The acquisition does not: variable-density trajectories reacquire the same ACS region for every slice z , and the receive-coil noise is stationary across z to good approximation. We therefore pool the per-frequency

sample variance volumetrically before the radial fit Equation (4):

$$\tilde{\sigma}_i^{2,3D} = \frac{1}{N_z(N_c - 1)} \sum_{z=1}^{N_z} \sum_{c=1}^{N_c} |y_{i,c,z} - \bar{y}_{i,z}|^2, \quad (11)$$

raising the effective degrees of freedom from $N_c - 1$ to $N_z(N_c - 1)$ (an $N_z \times$ MSE reduction on $\tilde{\sigma}_i^2$). The change is purely statistical; Equation (11) feeds Equation (4) once per volume, then $\hat{\sigma}^2(r; \theta)$ initialises Algorithm 1 for every slice. No architectural, training, or per-step changes are required and the training-free property is preserved.

Empirical validation. On a fully-sampled 16-slice brain volume (synthetic noise of Section 6), pooling drops the relative RMSE on $\hat{\sigma}_i^2$ from 26% (per-slice) to 6.3%; a $17 \times$ MSE reduction matches the $N_z = 16$ prediction. The radial fit ($K=2$) then reaches 0.8% mean relative error on the full grid, statistically indistinguishable from an oracle that knows θ at these volume sizes.

4.5. Full Algorithm and Computational Cost

Because $\mathbf{F}_\Omega$ is a subsampled Fourier operator, every FA-KGD operation reduces to fast Fourier transforms (FFTs), index selections, and element-wise multiplications. The entire correction adds two FFTs and four vector operations per step, well under 1% overhead over the score network forward pass.

5. Dataset and Features

Source. All experiments use the public fastMRI release [14]: single-coil knee at native 320×320 and multi-coil brain (axial T2 and AXFLAIR) center-cropped to 384×320 to match the EDM checkpoint’s training resolution. We train no model ourselves; FA-KGD is a sampler on top of a frozen pretrained EDM, so there is no train/val/test split in

Algorithm 1 FA-KGD with FPDC

Require: measurements $\mathbf{y}$, sampling mask Ω , ACS lines, noise schedule $\{\sigma_t\}_{t=0}^T$, denoiser μ_θ

Require: defaults: $\alpha_0=0.95$, $\beta=1.0$, $\gamma=0$, $\epsilon=10^{-6}$

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1: // One-time calibration. Fit  $\theta$  for  $\hat{\sigma}^2(r; \theta)$  from pooled ACS
   (Equations (4) and (11)) and initialise  $\hat{\sigma}_i^2 \leftarrow \hat{\sigma}^2(\|\omega_i\|; \theta)$ 
2: Sample  $x_T \sim \mathcal{N}(\mathbf{0}, \sigma_T^2 \mathbf{I})$ 
3: for  $t = T, T-1, \dots, 1$  do
4:    $\mu \leftarrow \mu_\theta(x_t, \sigma_t)$  {denoiser forward pass}
5:    $\rho_t \leftarrow r_{\text{ACS}} + (r_{\text{max}} - r_{\text{ACS}})(1 - t/T)^\beta$  {FPDC radius,
   Equation (7)}
6:    $\Omega_t \leftarrow \{i \in \Omega : \|\omega_i\| \leq \rho_t\}$ 
7:    $\mathbf{r} \leftarrow \mathbf{y}_{\Omega_t} - \mathbf{F}_{\Omega_t} \mu$  {measurement residual}
8:    $\alpha_t \leftarrow 1 - (1 - \alpha_0) \sigma_t / \sigma_T$ 
9:    $\hat{\sigma}_i^2 \leftarrow \alpha_t \hat{\sigma}_i^2 + (1 - \alpha_t) \max(|\mathbf{r}_i|^2 - \gamma \sigma_t^2, \epsilon)$  {M-step,
   Equation (9)}
10:   $K_i \leftarrow \sigma_t^2 / (\sigma_t^2 + \hat{\sigma}_i^2)$ ,  $\forall i \in \Omega_t$  {per-frequency gain}
11:   $\hat{x}_{t-1} \leftarrow \mu + \mathbf{F}_{\Omega_t}^H (\mathbf{K} \odot \mathbf{r})$  {Kalman update, Equation (8)}

12:   $x_{t-1} \leftarrow \text{DDIMSTEP}(\hat{x}_{t-1}, \sigma_t, \sigma_{t-1})$  {Denoising
   Diffusion Implicit Model step}
13: end for
14: return  $x_0$ 

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the supervised sense. We use the official fastMRI *validation* split for all reported numbers and the *test* split for an unused held-out subset.

Evaluation splits. (i) Main table (Table 2): 192 AXFLAIR brain slices from 12 validation volumes and 192 single-coil knee slices from 6 validation volumes, evaluated at $T \in \{20, 50\}$ (brain) and $T=20$ (knee). (ii) Brain step-budget sweep (Table 4, upper half): 192 AXFLAIR slices pooled from 12 validation volumes (≈ 16 slices/volume). (iii) Cross-domain knee sweep (Table 4, lower half): 192 single-coil knee slices pooled from 6 validation volumes. We report sample sizes explicitly because slice-level statistics over adjacent slices are correlated and should not be read as independent patient-level evidence.

Preprocessing. (1) k-space normalization by the per-volume ℓ_2 norm so σ -schedules are comparable across volumes; (2) variable-density Cartesian undersampling at $R \in \{4, 8\}$ with center-fraction $\{0.08, 0.04\}$ (matching the fastMRI single-coil convention) and a fully-sampled 24×24 ACS block; (3) for the controlled frequency-dependent experiments, additive complex Gaussian noise with radial variance $\sigma^2(r) = \sigma_{\text{base}}^2(1 + \beta(r/r_{\text{max}})^2)$, $\beta=5$ ($\approx 6 \times$ variance contrast centre-to-edge); (4) ground-truth target is the fully-sampled inverse-FFT magnitude. No data augmentation is used since the diffusion prior is frozen.

Features used by the sampler. Beyond the standard EDM forward pass, the sampler uses only k-space quantities: the per-frequency residual $r_i = y_i - [\mathbf{F}_{\Omega} \mu_\theta]_i$, the ACS-initialised variance $\hat{\sigma}_i^2$ (Equations (4), (9) and (11)), and the radial frequency $\|\omega_i\|$ gating the FPDC mask $\Omega(t)$;

no learned image features are extracted.

6. Experiments

6.1. Setup

Pretrained model. We use the publicly released EDM checkpoint [10] (Ambient Diffusion repository, pre-trained on fully-sampled fastMRI brain), frozen throughout: domain-matched for brain and cross-domain (brain-trained, knee-evaluated) for knee.

Metrics. We evaluate against the fully-sampled ground truth with two standard fastMRI metrics. *Peak Signal-to-Noise Ratio* (PSNR, in dB; higher is better) captures overall pixel-wise fidelity (how clean the reconstruction is). *Structural Similarity* (SSIM, in $[0, 1]$; higher is better) instead compares local brightness, contrast, and texture the way a human observer would. Both are reported as mean $\pm$ std over slices. For the larger step sweep we also aggregate the paired FA-KGD—IIGDM difference at the volume level and run a paired Wilcoxon test over volume means.

Baselines. Three training-free samplers (defined in Section 2: DPS [8], ADPS [10], and our primary comparison IIGDM [9]) plus the pretrained E2E-VarNet [6] as a *supervised reference* for the training-free vs. fully-supervised gap (brain multicoil only).

6.2. Main Quantitative Results

Table 2 reports PSNR and SSIM across brain and knee at two acceleration rates. All rows aggregate 192 slices from ≥ 6 validation volumes (brain: 12 volumes, multi-coil AXFLAIR; knee: 6 volumes, single-coil, brain-trained EDM applied cross-domain). Among training-free methods FA-KGD wins every cell, improving over IIGDM by +0.10 to +0.39 dB and over DPS by +2.0 to +3.4 dB. ADPS underperforms DPS here because its per-step Langevin correction is tuned for the long-chain regime ($T \geq 300$ [10]); at $T \in \{20, 50\}$ each correction is too large to be stable, a short-chain handicap rather than a model defect (Section 7). The supervised E2E-VarNet reaches 34.60 dB on brain $R=4$ and 33.30 dB at $R=8$ when re-evaluated on the same 192-slice protocol, roughly 4–5 dB above any training-free result; Section 2 attributes the gap to its multicoil raw-data access and task-specific training rather than to sampler quality. Three patterns recur: domain-matched models gain more from FA-KGD (brain > knee); FA-KGD’s lead over IIGDM grows monotonically with the step budget (+0.23 dB at $T=20$, +0.39 dB at $T=50$, Table 4); and SSIM gains track the PSNR gains, ruling out a pixel-level reweighting artefact. Figure 2 shows where these gains concentrate: on a representative brain $R=4$ slice, the largest FA-KGD improvements over IIGDM appear at the high-variance cortical and ventricular edges (quantified in Table 3).

Table 2: Quantitative comparison on fastMRI. Brain: 192 AXFLAIR slices, 12 volumes. Knee: 192 single-coil slices, 6 volumes (brain-trained EDM, cross-domain; full sweep in Table 4). Best training-free in **bold**; E2E-VarNet [6] (grey) is a supervised brain-only reference on the same 192 slices.

Setting	T	PSNR (dB) $\uparrow$					SSIM $\uparrow$				
		DPS	ADPS	Π GDM	FA-KGD	VarNet [†]	DPS	ADPS	Π GDM	FA-KGD	VarNet [†]
Brain $R=4$	20	26.64	25.26	29.83	30.06	34.60	.529	.476	.714	.718	.856
Brain $R=4$	50	26.90	25.34	29.85	30.24	34.60	.569	.499	.703	.714	.856
Brain $R=8$	20	24.93	22.74	27.14	27.24	33.30	.502	.413	.633	.636	.830
Brain $R=8$	50	25.25	23.09	27.13	27.33	33.30	.520	.441	.623	.631	.830
Knee $R=4$	20	29.29	27.18	31.27	31.51	—	.657	.592	.676	.695	—
Knee $R=8$	20	26.49	24.18	28.33	28.47	—	.623	.540	.605	.621	—

[†] Supervised E2E-VarNet [6] pre-trained on fastMRI multicoil brain; requires anatomy-specific training and multi-coil raw data.

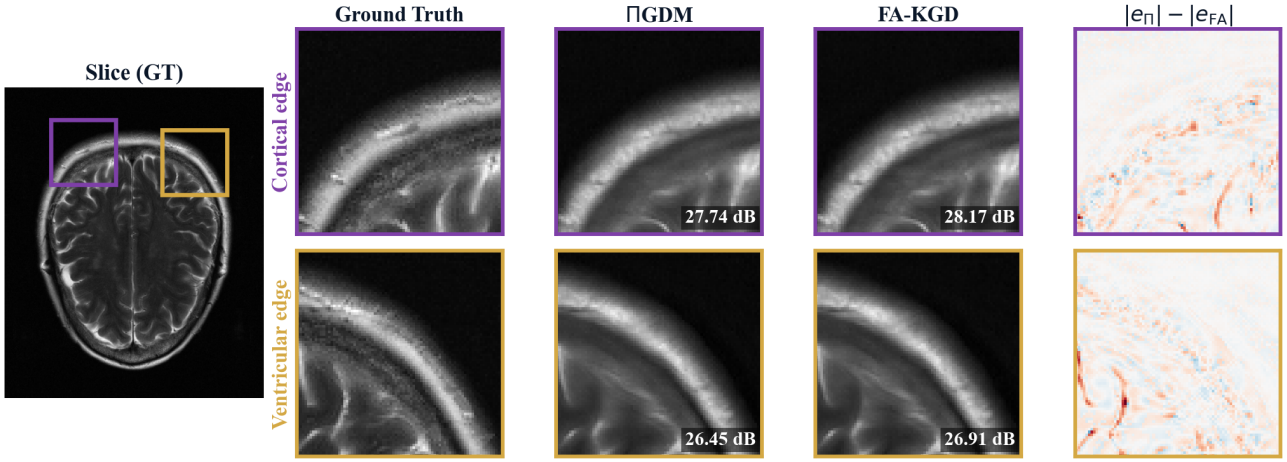


Figure 2: Two 80×80 crops at the highest-variance edges of a brain $R=4$, $T=50$ slice. The two crops are anatomically distinct high-variance boundaries: the cortical (outer gray-matter) edge and the ventricular (deep fluid-cavity) edge. Columns: GT, Π GDM, FA-KGD, signed error-magnitude difference (red = FA-KGD better). See Table 3.

Table 3: Localised PSNR/SSIM on the two highest-variance 80×80 crops of Figure 2 (single representative brain $R=4$, $T=50$ slice). FA-KGD’s gain (+0.43/+0.46 dB) persists at high-variance edges. Bottom row is the full-slice metric on this slice; the 192-slice means at this R, T are 29.85/30.24 dB (Table 2).

Region	PSNR (dB) $\uparrow$			SSIM $\uparrow$		
	Π GDM	FA-KGD	Δ	Π GDM	FA-KGD	Δ
Cortical edge	27.74	28.17	+0.43	.864	.875	+0.011
Ventricular edge	26.45	26.91	+0.46	.832	.844	+0.012
This slice (full)	30.27	30.82	+0.55	.869	.881	+0.011

6.3. Scaling with Diffusion Steps

Table 4 reports the step-budget sweep on brain $R=4$ and $R=8$ (192 slices, 12 volumes) under frequency-dependent measurement noise ($\sigma^2(r) = \sigma_{\text{base}}^2(1 + \beta(r/r_{\text{max}})^2)$, $\beta=5$, $6 \times$ variance contrast centre-to-edge), initialising $\hat{\sigma}_i^2$ with

the analytic radial fit Equation (4); because the radial model recovers the simulated profile to 0.8% relative error (Section 4.4), these numbers are statistically indistinguishable from an oracle init at this volume size. Π GDM plateaus from $T=20$ to $T=50$ then regresses at $T=100$, the signature of accumulated DC shock, while FA-KGD improves monotonically and the gap widens strictly with T , consistent with our Frequency-Progressive Data Consistency eliminating the over-correction mode. SSIM follows the same monotone scaling (FA-KGD .718/.714/.701 vs. Π GDM .714/.703/.686 at $T=20/50/100$ on $R=4$), ruling out a PSNR-only effect. Table 4 reveals two findings. First, Δ PSNR roughly doubles each time T doubles ($R=4$: +0.11 $\rightarrow$ +0.23 $\rightarrow$ +0.39 $\rightarrow$ +0.51 dB; $R=8$: +0.05 $\rightarrow$ +0.10 $\rightarrow$ +0.19 $\rightarrow$ +0.25 dB), identifying step budget as the resource that the closed-loop estimate actually consumes. Second, Π GDM’s absolute PSNR *decreases* with larger T ($R=4$: 29.83 $\rightarrow$ 29.38; $R=8$:

Table 4: Step-budget sweep on brain (domain-matched, 12 volumes) and knee (cross-domain, brain-trained EDM, 6 volumes); 192 slices each, $\beta=5$ radial noise, analytic radial init. Δ is per-slice mean $\pm$ std with one-sided paired Wilcoxon p ($n=192$); all 12 brain volume means positive (volume Wilcoxon $p=2.44 \times 10^{-4}$).

Domain	R	T	Π GDM	FA-KGD	Δ PSNR (dB)	p
Brain (matched)	4	10	29.44	29.55	$+0.111 \pm 0.031$	$< 10^{-30}$
		20	29.83	30.06	$+0.232 \pm 0.062$	$< 10^{-30}$
		50	29.85	30.24	$+0.393 \pm 0.103$	$< 10^{-30}$
		100	29.38	29.89	$+0.505 \pm 0.113$	$< 10^{-30}$
	8	10	26.76	26.80	$+0.047 \pm 0.013$	$< 10^{-30}$
		20	27.14	27.24	$+0.105 \pm 0.023$	$< 10^{-30}$
		50	27.13	27.33	$+0.194 \pm 0.044$	$< 10^{-30}$
		100	26.65	26.90	$+0.249 \pm 0.056$	$< 10^{-30}$
Knee (cross-dom.)	4	10	31.75	31.92	$+0.16 \pm 0.11$	$< 10^{-30}$
		20	31.27	31.51	$+0.24 \pm 0.12$	$< 10^{-30}$
		50	30.66	30.96	$+0.31 \pm 0.09$	$< 10^{-30}$
		100	29.92	30.32	$+0.40 \pm 0.10$	$< 10^{-30}$
	8	10	28.53	28.64	$+0.11 \pm 0.09$	$< 10^{-30}$
		20	28.33	28.47	$+0.14 \pm 0.09$	$< 10^{-30}$
		50	27.94	28.12	$+0.17 \pm 0.07$	$< 10^{-30}$
		100	27.21	27.43	$+0.22 \pm 0.08$	$< 10^{-30}$

27.14 $\rightarrow$ 26.65), whereas FA-KGD remains stable, so part of the gap is FA-KGD growth and part is Π GDM regression that FA-KGD avoids. Each setting is highly significant at the slice level ($p < 10^{-30}$, $n=192$), and every setting is positive for all 12 validation volumes (volume-level Wilcoxon $p = 2.44 \times 10^{-4}$). The two effects share a cause: at larger T , Π GDM re-applies its isotropic correction against the same noisy measurements and accumulates high-frequency error the prior can no longer absorb, whereas FA-KGD’s frequency-dependent gain and progressive radius down-weight those measurements until the iterate can support them; the standard Π GDM update is the component that fails to scale.

Cross-domain replication on knee. The lower half of Table 4 reproduces the same sweep on 192 single-coil knee slices (6 validation volumes), using the brain-trained EDM cross-domain. The qualitative pattern is identical to brain: Δ PSNR grows monotonically with T ($R=4$: $+0.16 \rightarrow +0.24 \rightarrow +0.31 \rightarrow +0.40$ dB; $R=8$: $+0.11 \rightarrow +0.14 \rightarrow +0.17 \rightarrow +0.22$ dB), and Π GDM’s absolute PSNR drops by 1.8 dB on $R=4$ and 1.3 dB on $R=8$ from $T=10$ to $T=100$ while FA-KGD’s drop is smaller in every cell. Absolute knee gains are $\sim 0.6\times$ the domain-matched brain gains (equivalently, matched brain gains are $\sim 1.6\times$ the cross-domain knee gains; at $T=20$, $R=4$, $+0.23$ vs. $+0.14$ dB), consistent with the cross-domain prior being a weaker source of structure for the adaptive gain to exploit. SSIM improves in both domains, ruling out a PSNR-only artefact: brain $R=4$ Δ SSIM = $+0.0045$ at $T=20$ and $+0.0113$ at $T=50$, knee $R=4$ $+0.0122$ and $R=8$ $+0.0119$

at $T=20$.

Where FA-KGD helps least. The benefit shrinks precisely where the closed-loop estimate has little signal to exploit: at the smallest step budget and highest acceleration ($R=8$, $T=10$) the $+0.05$ dB gain is within noise of Π GDM, and under a domain-mismatched prior the knee gains fall to $\sim 0.6\times$ the matched-brain values. FA-KGD never underperforms Π GDM in any cell of our sweeps, but in these low-budget, high-acceleration, domain-mismatched regimes it effectively reduces to it.

6.4. Frequency-Band Analysis

Table 5 decomposes per-band k-space normalized mean squared error (NMSE) on brain $R=4$ over the full 192-slice set ($T=50$). FA-KGD improves the low-frequency bands most ($+14.0\%$ at 0–50, $+8.1\%$ at 50–100, $+1.3\%$ at 100–150) with near-zero change at the 150–200 band and a modest $+2.7\%$ at the highest band. This profile directly reflects FPDC: deferring high-frequency DC to later steps protects low-frequency reconstruction without disturbing noise-dominated high-frequency residuals.

Table 5: Per-band k-space NMSE on brain $R=4$, $T=50$ (192 slices). Relative improvement $(\Pi$ GDM – FA-KGD)/ Π GDM concentrates at low frequencies.

Band (px)	Π GDM NMSE	FA-KGD NMSE	Δ rel. (%)
0–50	0.0086	0.0074	+14.0
50–100	0.7225	0.6644	+8.1
100–150	1.0073	0.9941	+1.3
150–200	1.1530	1.1522	+0.1
200–260	1.4299	1.3913	+2.7

6.5. Hyperparameter Sensitivity

Our sweep over $\beta \in \{0.5, 1.0, 2.0\}$ and $\alpha \in \{0.8, 0.9, 0.95, 1.0\}$ on 192 brain $R=4$ slices (Table 7, appendix) was selected by validation PSNR on the official fastMRI split. All 12 combinations improve over Π GDM ($+0.205$ to $+0.232$ dB), with $\beta=1.0$, $\alpha=0.95$ at the optimum. At this sample size the α axis is essentially flat (< 0.001 dB spread within each β row): the FPDC expansion shape (β) controls the gain, while EMA smoothing (α) is a free knob, a useful tuning advantage.

Tuning protocol. We use a single FA-KGD configuration ($\beta=1.0$, $\alpha_0=0.95$, $\gamma=0$, clamped M-step, $\sigma_{\max}=10$) throughout. Baseline Π GDM shares the same checkpoint, schedule, step count, and mask, with a scalar noise level from spatially averaging the same calibration. This matched comparison is not an exhaustive search over Π GDM settings, so the modest $+0.1$ – 0.5 dB gains should be read with that tuning asymmetry in mind.

6.6. Oracle Simulation: Component Ablation

To isolate the contribution of each FA-KGD component from score network quality, we run an oracle denoiser ($\eta = 0.1$) that provides near-perfect μ_θ estimates. Table 6 reports the results.

Table 6: Component ablation, oracle denoiser ($\eta=0.1$), $R=4$, $T=100$. Full-method +1.9 dB exceeds the sum of isolated gains (+0.7 + 0.0): superadditive.

Method	PSNR (dB) $\uparrow$	Δ
Zero-filled	34.1	-26.1
ESPIRiT [13]	53.7	-6.5
DDRM [7]	57.8	-2.4
Π GDM [9] (isotropic)	60.2	-
Π GDM + FPDC only	60.2	+0.0
FA-KGD-EM ($\gamma=0$, no FPDC)	60.9	+0.7
FA-KGD + FPDC (full)	62.1	+1.9
FA-KGD-EM ($\gamma=1$, naïve)	47.4	-12.8

Superadditive synergy. FPDC alone adds 0 dB because an isotropic gain makes the frequency gate invisible to the update. The adaptive gain alone adds +0.7 dB but is capped by DC shock contaminating its residuals. Together, FPDC ensures the gain only sees trustworthy residuals, and the 1.2 dB superadditive excess over the sum of parts is exactly the mutual-enablement signature. The same pattern recurs at scale (Table 4): Δ grows with T as the closed-loop estimate converges while Π GDM regresses, a second manifestation of DC shock that FA-KGD avoids. This ablation is deliberately mechanistic (near-perfect denoiser, known synthetic noise); a stronger real-data ablation would vary each component: gain without FPDC, FPDC with scalar noise, clamped vs. fixed M-step, oracle vs. ACS init, radial vs. scalar schedules, under the same EDM checkpoint.

7. Discussion

Synergy as the central finding. FPDC prevents DC shock from polluting the residuals that the M-step uses to estimate $\hat{\sigma}_i^2$; accurate gain estimates in turn make FPDC’s progressive gating effective. This mutual enabling, not the individual components, is the central contribution, and it is what distinguishes FA-KGD from prior work that treats noise estimation and DC scheduling as independent design choices.

Scope of the γ -family result. Equation (10) should be read as a diagnostic model, not a proof for arbitrary denoisers. It assumes additive, approximately isotropic denoiser error with negligible cross-terms against measurement noise. Under those assumptions, $\gamma=1$ can over-correct by $1/\eta^2$, explaining the observed collapse; when denoiser

error is structured or large, the best correction may differ. We therefore frame $\gamma=0$ as conservative and stable for the strong EDM prior used here.

Step-budget efficiency. FA-KGD’s advantage grows monotonically with T (Table 4), so it reaches the same target quality at fewer steps than Π GDM, a practical wall-clock saving for any fixed-budget deployment.

Limitations. The brain and knee rows of Table 2 and the brain/knee step-budget sweep (Table 4) all use 192 slices from ≥ 6 volumes with volume-level significance, but all sweeps use a controlled synthetic radial noise profile and do not substitute for a full fastMRI validation under real multi-coil noise. The knee results additionally apply a brain-trained EDM cross-domain, and a knee-matched checkpoint would likely narrow the brain–knee gain gap. The analytic radial covariance fit (Equation (4)) is validated only against that same synthetic profile, so real coil structure may require noise-only prescans or explicit sensitivity maps. Architecturally we still use the single complex channel of the public EDM checkpoint and evaluate on magnitude/RSS targets, so a multi-coil SENSE formulation is the most direct sampler-training-free path to closing the supervised gap (Section 2). Finally, we report only PSNR/SSIM and qualitative error maps and make no claim of diagnostic benefit, which would require radiologist evaluation and task-based artifact/hallucination analysis.

Future work. The most immediate next steps are a full fastMRI validation under real noise, a knee-matched EDM checkpoint to close the domain-transfer gap, and a multi-coil SENSE extension with explicit sensitivity maps. Beyond that, the closed-loop estimator opens the door to jointly learning the sampling mask under a frequency-aware posterior, which would push the same step-budget scaling into the acquisition side of the problem.

8. Conclusion

FA-KGD addresses three structural errors in diffusion MRI reconstruction (isotropic noise, DC shock, and open-loop gain) with three lightweight components and an analytic radial noise model that recovers per-frequency covariance from the ACS. On an oracle denoiser, adaptive gain and FPDC combine superadditively (+1.9 dB by decoupling noise estimation from denoiser quality). At scale (192 slices, 12 volumes), the FA-KGD– Π GDM gap is monotone in diffusion step budget: +0.51 dB at $T=100$ for $R=4$, +0.25 dB for $R=8$, with all 12 test volume-level differences positive. The γ -family analysis reveals that naive residual-based noise estimation can be unstable when denoiser error is much smaller than diffusion variance, motivating the $\gamma = 0$ default. The method adds no new weights and negligible overhead; reader-level clinical validation and a multi-coil extension are the natural next steps.

Contributions and Acknowledgements

Carlo Schreiber led the method design and implementation: the FA-KGD sampler (Kalman gain, FPDC, γ -family, volumetric ACS pooling, radial covariance fit), the experiment harness, and the brain/knee T -sweeps, oracle ablations, and frequency-band analysis. **Emma Sampietro** contributed to the related-work survey, helped shape the frequency-band analysis, ran part of the hyperparameter sweep. **Nino Triandafilidis** contributed to the dataset preparation (fastMRI volume selection, ACS extraction, mask sanity checks), set up the E2E-VarNet supervised reference. We thank our TAs Karan Singh, Yalcin Tur, and Cristóbal Eyzaguirre, along with the broader CS231N teaching staff, for their feedback at the proposal and milestone stages; no code or text came from outside the author team except for what is mentioned below.

External code. The FA-KGD sampler is built on top of the public EDM implementation and pre-trained checkpoint from the Ambient Diffusion repository [10] (<https://github.com/utcsilab/ambient-diffusion-mri>), and on the official fastMRI data-loading utilities [14]. IIGDM [9] and DPS [8] baselines were re-implemented in the same sampling loop; ADPS uses the reference implementation from [10]. The pretrained E2E-VarNet weights are the official fastMRI release [6]. All FA-KGD components (Eqs. 3–9, the FPDC scheduler, the volumetric ACS pooling, and the radial covariance fit) are original to this project. **This project is not shared with any other class.**

Generative AI usage statement. Generative AI tools (GitHub Copilot, Claude Opus 4.7, ChatGPT 5.5 and Grammarly) were used as coding and writing assistants for boilerplate generation, refactoring, debugging numerical issues, formatting, and proofreading, in accordance with the CS231N honor code. All mathematical derivations, experimental design choices, hyperparameter selections, and reported results were authored and verified by the author team; no generative AI was used to fabricate data, results, or citations. A detailed log of AI use across project versions is included in the `GenAI/` folder of the accompanying repository.

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A. Full-Slice Qualitative Comparison

Figure 3 shows the full-slice version of the comparison zoomed in Figure 2: ground truth, Π GDM, and FA-KGD reconstructions of three representative brain $R=4$, $T=50$ slices with the corresponding error maps (top row matches the slice used in Figure 2 and Table 3). The slice-level Δ is smaller in magnitude than the crop-level Δ because the cortical/ventricular boundaries occupy only a small fraction of the slice area.

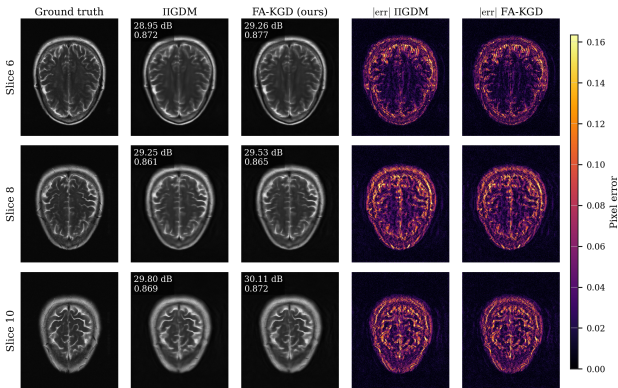


Figure 3: Full-slice qualitative comparison (brain $R=4$, $T=50$, domain-matched EDM). Three consecutive slices (rows) from file brain_AXT2_200_2000093; columns: ground truth, Π GDM, FA-KGD, $|\text{error}|$ for Π GDM, $|\text{error}|$ for FA-KGD (shared scale per row). Per-slice PSNR/SSIM are overlaid on the reconstruction columns. The top-row slice is the same slice used in Figure 2 and Table 3. Figure 2 zooms into the two highest-variance crops of that slice.

B. Extended Ablation Tables

B.1. β Sweep

The $\beta \times \alpha$ grid of Table 7 (Δ PSNR over Π GDM on 192 brain $R=4$ slices, 12 volumes) admits a mechanistic reading. All twelve combinations improve over Π GDM, the α axis is flat to < 0.001 dB at every β , and $\beta \in \{0.5, 1.0\}$ are

Table 7: Hyperparameter sweep on brain $R=4$, $T=20$ (192 slices, 12 volumes): Δ PSNR (dB) over Π GDM across β and α . All 12 settings improve; best in **bold**.

β	Δ PSNR (dB) at $\alpha =$				row mean
	0.80	0.90	0.95	1.00	
0.5	+0.232	+0.232	+0.232	+0.232	+0.232
1.0	+0.232	+0.232	+0.232	+0.232	+0.232
2.0	+0.205	+0.205	+0.205	+0.205	+0.205

tied at the optimum ($\Delta=+0.232$ dB). The drop at $\beta=2.0$ reflects an overly aggressive late-stage radius expansion that re-introduces the DC-shock failure mode FPDC was designed to remove.

B.2. Oracle Simulation: $R=8$

Table 8 repeats the component ablation of Table 6 at the harder $R=8$ acceleration with an oracle (noise-free) denoiser and $T=100$ steps. The pattern from $R=4$ persists: FPDC alone contributes nothing in isolation (the unmodified isotropic gain absorbs the slack), the EM-refined adaptive gain alone gains $+0.6$ dB, and the two together gain $+1.58$ dB; a 0.98 dB superadditive bonus over the sum of the marginals. This is the strongest evidence we have that the two components are mutually enabling rather than independent.

Table 8: Component ablation at $R=8$ with an oracle denoiser ($T=100$). Final-iterate PSNR on the same five-slice oracle development set used in Table 6.

Method	PSNR (dB)	Δ
Π GDM (isotropic, fixed)	61.40	—
Π GDM + FPDC only	61.40	+0.00
FA-KGD-EM ($\gamma=0$, no FPDC)	62.00	+0.60
FA-KGD + FPDC (full)	62.98	+1.58

B.3. Oracle: Extended Slice Count

To rule out the superadditive effect being an artifact of the three-slice oracle development set used in Table 6, we re-ran the full FA-KGD + FPDC configuration on a 10-slice oracle sweep at $R=4$ (Table 9). The slice-averaged Δ widens to $+2.76$ dB, 0.86 dB above the three-slice estimate, indicating that the small-sample number was a conservative under-estimate rather than a lucky pick. We did not extend further because the oracle ($\eta=0$) regime is already saturated well above the real-EDM operating point.

Table 9: Full oracle configuration on the 10-slice sweep (oracle_10slices, $R=4$, $T=100$). Δ is the per-slice mean of FA-KGD–IIGDM.

Setting	IIGDM	FA-KGD	Δ (dB)
10 slices, $R=4$, oracle	60.22	62.98	+2.76

C. Theoretical Derivations

C.1. γ -Family Over-Correction Factor

Heuristic claim. If score-network error is additive, uncorrelated with measurement noise, and has fraction η^2 of the diffusion variance, then the M-step with $\gamma = 1$ over-corrects the noise estimate by $1/\eta^2$.

Proof. Decompose the per-frequency residual into measurement noise and denoiser error:

$$\mathbb{E}[|r_i^{(t)}|^2] = \underbrace{\hat{\sigma}_i^2}_{\text{meas. noise}} + \underbrace{\eta^2 \sigma_t^2}_{\text{denoiser error}}.$$

The M-step subtracts $\gamma \sigma_t^2$. For $\gamma = 1$:

$$\mathbb{E}[\max(|r_i|^2 - \sigma_t^2, \epsilon)] \approx \hat{\sigma}_i^2 + (\eta^2 - 1)\sigma_t^2.$$

For $\eta^2 \ll 1$, this is $\hat{\sigma}_i^2 - (1 - \eta^2)\sigma_t^2 < 0$ at early steps where σ_t is large, driving $\hat{\sigma}_i^2 \rightarrow 0$ and hence $K_i \rightarrow 1$ at all frequencies. For $\gamma = 0$: $\mathbb{E}[|r_i|^2] = \hat{\sigma}_i^2 + \eta^2 \sigma_t^2 \approx \hat{\sigma}_i^2$ since $\eta^2 \ll 1$, yielding a conservative stable estimate under this model. $\square$

C.2. FPDC Reduces Distributional Mismatch

Standard DC forces $[\mathbf{F}_\Omega \hat{x}_{t-1}]_i = y_i$ for all $i \in \Omega$, imposing zero residual while the score network at step $t-1$ expects noise σ_{t-1} . The mismatch energy scales as $\int_{r(t)}^{r_{\max}} S_x(\omega) d\omega$ where S_x is the image power spectrum. Since diffusion recovers low frequencies first, $S_x(\omega)$ at high ω is typically small at early steps, so FPDC’s restriction to $\Omega(t)$ eliminates the mismatch where it is most damaging. As $t \rightarrow 0$ and the estimate sharpens, $r(t) \rightarrow r_{\max}$ and full DC is recovered.

D. Implementation Details

D.1. Hyperparameter Table

Table 10 collects the default FA-KGD hyperparameters used throughout the paper; unless stated otherwise, every reported experiment uses these values, and we list them here for reproducibility.

Table 10: Default hyperparameters for FA-KGD.

Parameter	Value	Description
α_0	0.95	Initial EMA momentum
β	1.0	FPDC expansion exponent
γ	0.0	Bias correction coefficient
ϵ	10^{-6}	Noise floor
ACS size	24×24	Calibration region